

Madeleine Tang Schmidlin (Adams)

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SUMMARY

Experienced process development engineer with expertise in high-temperature materials conversion, experimental design, and process risk assessment. Strong record of advancing understanding of material-property-process relationships and applications to research and manufacturing challenges. Highly effective in cross-functional environments, partnering with characterization scientists, researchers, plant engineers, and technicians to drive technical insight and practical outcomes. Recognized for technical leadership through invited presentations, mentoring, two filed patents, and 10+ internal technical reports.

EXPERIENCE

Corning Incorporated

Staff Scientist – Melt Process Development, Gorilla Glass (2025 – Present) Corning, NY

- Led melt process development for advanced glass and glass-ceramic systems for mobile consumer electronics. Translated process-structure-property understanding into manufacturable solutions aimed at improved product quality, reduced equipment risk, and informing development strategy.
- Demonstrated that alternative chemical refining agents improved quality in glass-ceramics, eliminating measurable defects while offering potential cost, sustainability, and supply-chain benefits through reduced reliance on 3TG conflict minerals (patent pending). Presented as an invited talk at the American Ceramics Society (ACerS) 2026 Spring Meeting.
- Evaluated novel glass and glass-ceramic compositions against critical process contact points, aimed at mitigating equipment degradation and protecting core assets.
- Applied process-property understanding to establish operating windows reducing inclusion defects by 90% while achieving target material performance and maintaining safe, reliable operation (details protected under Trade Secret).
- Planned and executed experiments to evaluate new equipment for high-temperature glass processes. Assessed effects on product homogeneity and quality, generating insights to guide future design decisions.
- Supervised an intern on a melt process development study. Defined project scope, provided technical direction, and supported analysis of process effects on key material properties.

Process Engineer | Sr. Process Engineer, Corporate Engineering (2019 – 2025) Corning, NY

- Provided process engineering support across Corning businesses. Led equipment startup, manufacturing scale-up, process development, and troubleshooting initiatives.
- Assessed multiple glass and glass-ceramic compositions for viability on a sustainable all-electric melting platform. Conducted screening experiments on an induction melter system (patent pending – WO2025174508) and developed a Computational Fluid Dynamics (CFD) model using CelSian GTM-X software for the same system.
- Mitigated quality risk across \$2.9B of Corning Optical Communications revenue by standardizing built-in quality control protocols across manufacturing facilities worldwide.
- Accelerated pharmaceutical vial output by launching production of new product for \$20.5M Pfizer order as SKU-change lead. Managed equipment change-over, process development and qualification. Developed plans to minimize downtime.
- Led root cause analyses to determine causes and possible solutions to vial cosmetic defects through visual inspection, chemical analysis, and experimentation, demonstrating reduced defects on manufacturing line.
- Executed Factory Acceptance Testing (FAT) for \$4M of glass vial converting equipment, resolving critical vendor-side defects prior to shipment to eliminate downtime risks and protect the timeline for a major high-volume expansion.
- Performed glass forming line debug, documentation, validation, and start-up to produce COVID-19 vaccine vials on an accelerated timeline in support of \$204M government contract (BARDA Operation Warp Speed initiative).

EDUCATION

Cornell University (2022 – 2023) Ithaca, NY

Master of Engineering in Materials Science and Engineering. Cum Laude.

Oregon State University (2015 – 2019) Corvallis, OR

Bachelor of Science in Chemical Engineering. Cum Laude.

SKILLS

Process Development | High-Temperature Materials Conversion | Mandarin Chinese (Professional Proficiency) | Statistics | Experiment Design | Six Sigma Green Belt (DMAIC) | JMP | Python | FactSage Thermodynamic Modeling | CelSian GTM-X

OTHER ACTIVITIES

- Invited Speaker, American Ceramics Society Glass Manufacturing Symposium (2026)
- Treasurer, Society of Women Engineers – Twin Tiers Section (2020–2022)
- Completed Portland Marathon (2026)